



Stony Brook University

DATA ANALYSIS

AMS 572

Summarizing and Exploring Data

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Content

- ▶ Types of variables
- ▶ Measures of location
- ▶ Measures of spread, shape
- ▶ Data displays



Types of variables

- ▶ A *variable* is a quantity that may vary from object to object
- ▶ A *sample* or *data set* is a collection of values of one or more variables.
- ▶ Quantitative variable intrinsically numerical
e.g. age, height, counts
- ▶ Qualitative (categorical) - intrinsically nonnumerical
e.g. gender, province, country

.....



Types of variables

- ▶ Qualitative (categorical) - intrinsically nonnumerical
 - ▶ Binary, dichotomous
e.g., alive/dead, female/male
 - ▶ Ordinal - natural ordering
e.g., diagnosis (certain, probable, unlikely, ...)
e.g., attitude (strongly agree, agree, neutral, ...)
 - ▶ Nominal - no natural ordering
e.g., religion, race
- ▶ In recording qualitative data, numerical values may be assigned



Measures of Location

- ▶ (Arithmetic) Mean
- ▶ Percentiles
- ▶ Median
- ▶ Mode
- ▶ Geometric mean

Arithmetic mean

- ▶ Data:

$$x_1, x_2, \dots, x_n$$

- ▶ Mean:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

- ▶ Example: Duration of hospital stay in days:

$$x_1 = 5, x_2 = 10, x_3 = 6, x_4 = 11$$

Mean:

$$\bar{x} = 8$$

...

.....



Properties of Mean

► Let c be any constant

► If

$$y_i = x_i + c \text{ for } i = 1, 2, 3, \dots, n,$$

then

$$\bar{y} = \bar{x} + c$$

► If

$$y_i = cx_i \text{ for } i = 1, 2, 3, \dots, n,$$

then

$$\bar{y} = c\bar{x}$$

Properties of Mean - Example

- ▶ A sample of birth weights in a hospital found

$$\bar{y} = 3166.9 \text{ grams}$$

- ▶ 1 oz = 28.35 g
- ▶ Therefore the mean in ozs. is

$$\bar{x} = \frac{\bar{y}}{28.35} = 111.7$$

.....

Order Statistics

- ▶ Data: $x_1, x_2, \dots, x_n$
- ▶ Order data from smallest to largest

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$$

- ▶ $x_{(1)}, x_{(2)}, \dots, x_{(n)}$ are *order statistics*
- ▶ Note

$$x_{(1)} = \min\{x_1, x_2, \dots, x_n\}$$

$$x_{(n)} = \max\{x_1, x_2, \dots, x_n\}$$

- ▶ Example: Duration of hospital stay in days:

$$x_1 = 5, x_2 = 10, x_3 = 6, x_4 = 11$$

Order statistics:

$$x_{(1)} = 5, x_{(2)} = 6, x_{(3)} = 10, x_{(4)} = 11$$



Theorem: Let $X_{(1)} \leq \dots \leq X_{(n)}$ denote the order statistics of a random sample, $X_1, \dots, X_n$ from a random population with c.d.f. $F(\cdot)$ and p.d.f. $f(\cdot)$. The p.d.f. of $X_{(k)}$ is

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$$

Percentiles

- ▶ Intuitive definition: the x *percentile* is such that $x\%$ of the observations are less than that value
- ▶ Also known as sample *quantile*
- ▶ The $(p \times 100)^{th}$ percentile of a sample

$$\hat{\zeta}_p = \begin{cases} y_{(np+p)} & \text{if } np + p \text{ is an integer} \\ \{y_{(\lfloor np+p \rfloor)} + y_{(\lceil np+p \rceil)}\}/2 & \text{otherwise} \end{cases}$$

for $0 < p < 1$

Percentiles: General form

- ▶ General form (Hyndman and Fan, *Am Stat* 1996)

$$\hat{\zeta}_p = (1 - \gamma)y_{(j)} + \gamma y_{(j+1)}$$

where $j = \lfloor pn + k \rfloor$ for some $k \in \mathbb{R}$ and $0 \leq \gamma \leq 1$.

- ▶ Your textbook

$$\hat{\zeta}_p = \begin{cases} y_{(np+p)} & \text{if } np + p \text{ is an int} \\ y_{(m)} + [np + p - m](y_{(m+1)} - y_{(m)}) & \text{otherwise} \end{cases}$$

where $m = \lfloor np + p \rfloor$



Example

- ▶ In R, there are nine different quantile definitions (argument type)

```
> x <- 1:278
```

```
> quantile(x,.75,type=1)
```

```
75%
```

```
209
```

Median

- ▶ The sample median is the 50th percentile

$$\hat{\zeta}_{.5} = \begin{cases} y_{(\frac{n+1}{2})} & \text{if } n \text{ is odd} \\ \{y_{(n/2)} + y_{(n/2+1)}\}/2 & \text{if } n \text{ is even} \end{cases}$$

for $0 < p < 1$

- ▶ Example: Duration of hospital stay in days:

$$x_1 = 5, x_2 = 10, x_3 = 6, x_4 = 11$$

Median:

$$\hat{\zeta}_{.5} = 8$$



Mode

- ▶ The mode is the most frequently occurring value in the data set
- ▶ E.g., if

$$x_1 = 5, x_2 = 11, x_3 = 6, x_4 = 11$$

then mode is 11

Geometric Mean

- ▶ Data: $x_1, x_2, \dots, x_n$
- ▶ The geometric mean of x is

$$\bar{x}_g = (x_1 x_2 \cdots x_n)^{1/n}$$

- ▶ Eg, suppose $x_1 = 10$ and $x_2 = 0.1$. Then $\bar{x}_g = 1$



Comments

- ▶ Mean is most often used measure
- ▶ Median is better if there are influential observations (more robust to extreme values)
- ▶ Mode rarely used (exception: nominal data)



Measures of Spread, Shape

- ▶ Range
- ▶ Variance and standard deviation
- ▶ Interquartile range
- ▶ Skewness, Kurtosis

Range

- ▶ Range:

$$r_a = x_{(n)} - x_{(1)}$$

- ▶ Easy to calculate
- ▶ Sensitive to unusual observations (outliers)
- ▶ Usually, the larger n is, the larger r_a
- ▶ A rough estimate of $\sigma = r_a/4$

Sample Variance and Standard Deviation

- ▶ Want to measure deviation from mean
- ▶ Sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)$$

- ▶ Sample standard deviation

$$s = \sqrt{s^2}$$



Sample Variance and Standard Deviation

- ▶ An alternative form of the sample variance is

$$\hat{\sigma}_1^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

- ▶ We have shown that s^2 is unbiased for population variance σ^2 , however

$$E(\hat{\sigma}_1^2) = \sigma^2 - \frac{\sigma^2}{n}$$

Sample Standard Deviation

- ▶ The units of s are the same as the units of x_i
- ▶ If s is large, the data are spread over a wide range
- ▶ If c is a constant and

$$y_i = x_i + c,$$

then

$$s_y = s_x$$

- ▶ If

$$y_i = cx_i$$

then

$$s_y = cs_x$$

.....



Some approximations

- ▶ The interval $\bar{x} \pm s$ will contain approx 68% of the observations
- ▶ The interval $\bar{x} \pm 2s$ will contain approx 95% of the observations
- ▶ Approx s by

$$s \approx \frac{\hat{\zeta}_{.75} - \hat{\zeta}_{.25}}{1.35}$$

- ▶ Note

$$\hat{\zeta}_{.75} - \hat{\zeta}_{.25}$$

is called *interquartile range*



Symmetry and Skewness

- ▶ Informally, define *symmetry* to indicate having a uniform or even distribution about the mean
- ▶ If a distribution is symmetric,

$$\text{mean} = \text{median}$$

- ▶ Data sets that are not symmetric are said to be *skewed*
- ▶ *Skewness* is a measurement of the degree to which a data set is skewed

Skewness

- ▶ Define r th sample moment about the mean

$$m_r = \frac{\sum_i (y_i - \bar{y})^r}{n} \text{ for } r = 1, 2, 3, \dots$$

- ▶ Definition of sample skewness:

$$a_3 = \frac{\sum_i (y_i - \bar{y})^3 / n}{\{\sum_i (y_i - \bar{y})^2 / (n - 1)\}^{3/2}}$$

- ▶ $a_3 > 0$ indicates skewness to the right

..

Kurtosis

- ▶ *Kurtosis* is a measure of the flatness or peakedness of a distribution; degree of archedness; thickness of tails
- ▶ Definition of *sample* kurtosis:

$$a_4 = \frac{\sum_i (y_i - \bar{y})^4 / n}{\{\sum_i (y_i - \bar{y})^2 / (n - 1)\}^2}$$

- ▶ $a_4 > 3$ indicates the distribution has heavier tails than the normal distribution.
- ▶ In R, skewness and kurtosis can be computed using functions `skewness()` and `kurtosis()` from `library(e1071)`.

Data display

- ▶ Simplest form is a line listing
- ▶ A *frequency table* gives the frequency of observations within a set of ordered intervals
- ▶ Intervals should be mutually exclusive and exhaustive
- ▶ 8 to 10 intervals is usually sufficient
- ▶ With the exception of the end intervals, the length of the intervals should be constant



Frequency Table - Example

Blood Pressure	Pop1	Pop2	Pop3
< 106	218	4	23
106-114	272	23	132
116-124	337	49	290
126-134	362	33	347
136-144	302	41	346
146-154	261	38	202
156-164	166	23	109
> 164	314	52	112
Total	2232	263	1561



Frequency Tables

- ▶ Table on previous slide example of *empirical frequency distribution*
- ▶ Difficult to compare blood pressure distributions due to different sample sizes
- ▶ Divide by sample size to get *empirical relative frequency distribution*



ERFD - Example

Blood Pressure	Pop1	Pop2	Pop3
< 106	0.098	0.015	0.015
106-114	0.122	0.087	0.085
116-124	0.151	0.186	0.186
126-134	0.162	0.125	0.222
136-144	0.135	0.156	0.222
146-154	0.117	0.144	0.129
156-164	0.074	0.087	0.070
> 164	0.141	0.198	0.072
Total	2232	263	1561

Graphs

- ▶ Histogram
- ▶ Box plot
- ▶ Trellis/conditional plots

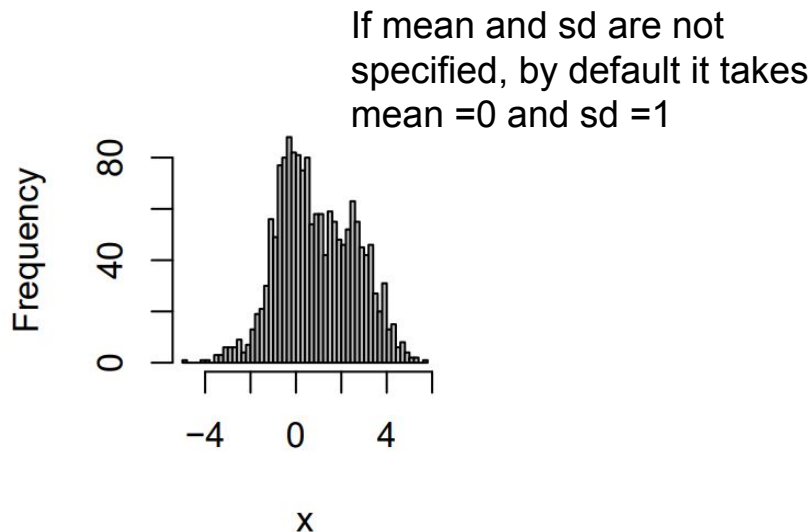
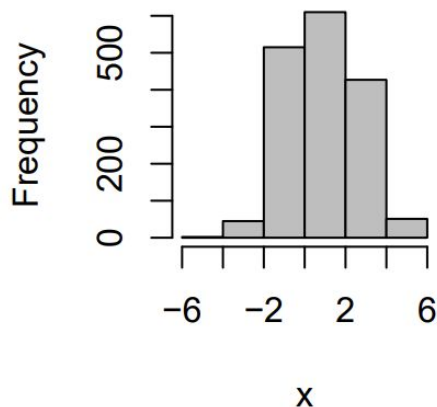


Histogram

- ▶ Data are divided into intervals as in a frequency table
- ▶ A histogram is a bar graph with the area of each bar equal to the relative frequency in the interval.
- ▶ Can compare histograms from samples of different size
- ▶ Intervals need not be the same width
- ▶ Beware effect of choice of interval width



```
> x <- c(rnorm(50,-2.5),rnorm(1000),rnorm(600,2.7))  
> hist(x,breaks=5,col="gray",xlab="x",main="")  
> hist(x,breaks=50,col="gray",xlab="x",main="")
```

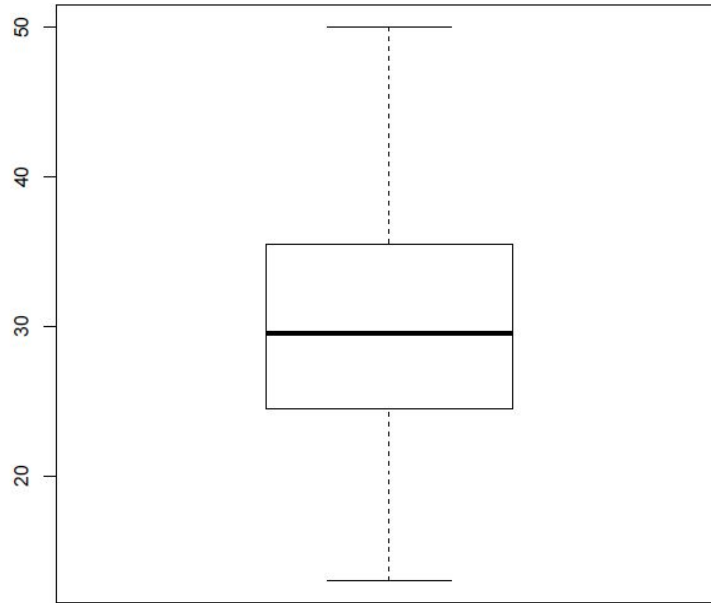


Box plot

- ▶ The top of the box is the 75th percentile ($\hat{\zeta}_{.75}$); the bottom is the 25th percentile ($\hat{\zeta}_{.25}$)
- ▶ A line through the box is drawn at the median
- ▶ The lines extending out of the box (*whiskers*) may extend to
 - ▶ the 90th and 10th percentiles
 - ▶ the largest and smallest values
 - ▶ largest observation $\leq \hat{\zeta}_{.75} + 1.5 \times \text{IQR}$;
smallest observation $\geq \hat{\zeta}_{.25} - 1.5 \times \text{IQR}$
- ▶ Data beyond whiskers may be plotted individually

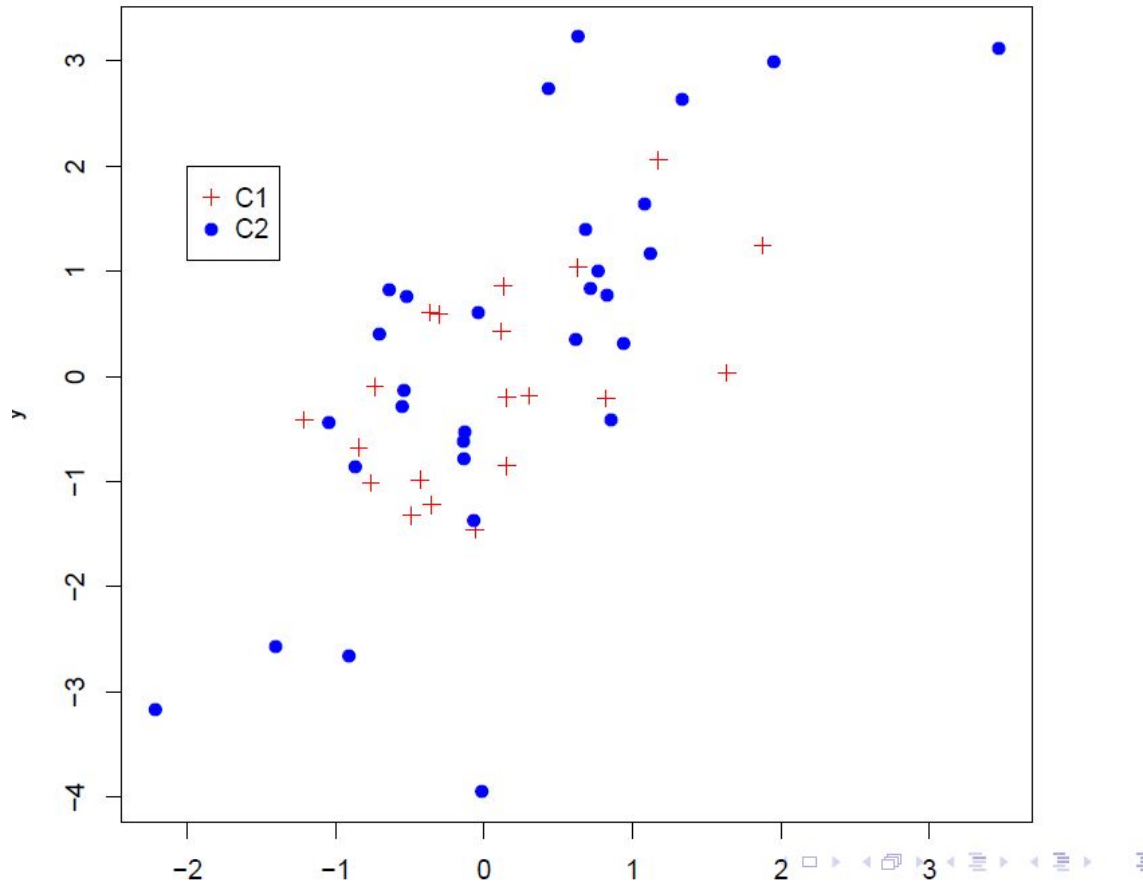
Boxplot Example

```
> boxplot(mileage)
```



Multivariate plots

- ▶ Describe relationships/associations between more than one variable
- ▶ Scatterplots
 - ▶ Simple for two variables
 - ▶ Add color, symbols for > 2 variables



**FAR
BEYOND**



```
> x <- rnorm(50)
> y <- x+rnorm(50)
> id <- sample(1:50,size=20)
> plot(x,y,type="n")
> points(x[id],y[id],col="red",pch=3)
> points(x[-id],y[-id],col="blue",pch=19)
> legend(-2,2,c("C1","C2"),col=c("red","blue"),pch=c(3,19))
```

Tables or graphs?

- ▶ Tables best suited for looking up specific information
- ▶ Graphs better for perceiving trends, making comparisons and predictions

Read Chapter 7